

Machine Learning for Business

QCM Exam

March 31, 2026

1. Machine learning primarily enables algorithms to:
 - (a) Replace all business managers
 - (b) Learn patterns and tasks directly from data
 - (c) Eliminate uncertainty from decisions
 - (d) Work only with fully labeled datasets
2. Why is probability necessary in machine learning ?
 - (a) Because outcomes are fixed once inputs are known
 - (b) Because models should avoid randomness completely
 - (c) Because identical inputs can still lead to different outputs
 - (d) Because all datasets are normally distributed
3. If the data set is represented by a table, and each row is an observation, the columns correspond to
 - (a) a random variable
 - (b) an event
 - (c) a probability
 - (d) an expectation
4. Assume that the data set is such that is row is an observation, and X is a column. Suppose that X is a column of a data set, whose rows are observations. The event $X \leq 10$ is
 - (a) collection of rows where $X \leq 10$
 - (b) collection of rows where $X > 10$
 - (c) the number of rows where $X \leq 10$
 - (d) the number of columns where $X > 10$
5. Two random variables X and Y have both zero mean and variance respectively $\sigma_X^2 = E[X^2] = 4$ and $\sigma_Y^2 = E[Y^2] = 5$ and they are both Gaussian. Choose the event which is the more likely to happen:
 - (a) $|X| > 0.2$
 - (b) $|Y| > 0.2$
6. Suppose that the one sampled t-test with the null hypothesis that the mean of a column X is 10 gave p-value 0.01. Can we reject the null hypothesis with a confidence level of 95% ?
 - (a) Yes
 - (b) No
7. We fitted a linear regression $y = w_0 + w_1x_1 + w_2x_2$ to data and the the confidence interval for w_2 is $[-1, 1]$ with probability 98%. Can we reject the null hypothesis that the true w_2 is zero, with the confidence level 98% ?

- (a) Yes
 - (b) No
8. We applied two linear regression models to two independent but identically distributed data sets D_1 and D_2 and computed the the square of the prediction error E_1 and E_2 . We checked with two sampled t-test if E_1 and E_2 have the same mean. The p-value was 0.01. Can we reject the hypothesis that the two models have the same performance with confidence level 95% ?
- (a) Yes
 - (b) No
9. A discrete random variable can take:
- (a) Any real number value
 - (b) Only finite values (e.g., 0, 1, 2, 3, ...)
 - (c) Only negative values
 - (d) Continuous values within an interval
10. The Central Limit Theorem implies that:
- (a) Individual observations are normally distributed
 - (b) The sample mean is approximately normal for large samples, regardless of the original distribution
 - (c) Variance increases with sample size
 - (d) All business data follows a normal distribution
11. Why can't we use linear regression directly for binary classification problems?
- (a) Linear regression is too slow
 - (b) Predictions may fall outside the $[0, 1]$ range
 - (c) Linear regression requires more data
 - (d) Linear regression cannot handle multiple variables
12. In gradient descent, the learning rate controls:
- (a) The direction of parameter updates
 - (b) The size of each step toward the minimum
 - (c) The number of iterations
 - (d) The final value of the loss function